Comite de Suivi

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Contents

1 Introduction

The objective of the present report is to provide an overview of the current status of the research projects pursued over the past year. The work I have conducted so far has focused on aspects of quantum cosmology in four dimensions, as well as on well-defined lower-dimensional toy models. Although the tools employed in these two settings are quite different, I believe that they share a common feature: they allow us to sharpen our understanding of the issues and features associated with quantizing a theory of gravity.

In the context of cosmology, the problem of quantizing gravity in Einstein's theory is nicely encapsulated by the wavefunction of the universe obeyinh the Wheeler-DeWitt equation. In 1983 [1], Hartle and Hawking postulated that the wavefunction of the universe should be computed as a path integral over compact geometries in the Euclidean past,

$$\Psi_{\mathrm{N.B}}[h_{ij}, \phi_0] = \int^{g_{ij}|_{\partial\mathcal{M}}=h_{ij}} \mathrm{Dg} \int^{\phi|_{\partial\mathcal{M}}=\phi_0} \mathrm{D}\phi\, \mathrm{e}^{-\mathcal{S}_{\mathrm{E}}[\mathrm{g},\phi]}\,. \tag{1.1}$$

The path integral is taken over all compact Euclidean geometries with a single boundary $\partial\mathcal{M}$ on which the induced metric and matter field are respectively h_{ij} and ϕ_0. Owing to the complexity of such an object, the path integral cannot be computed exactly, and one must resort to a series of truncations and approximations in order to evaluate it.

A well-known result in the context of single-field slow-roll inflation is obtained from the WKB evaluation of the minisuperspace truncation of the path integral. The modulus of the no-boundary instanton wavefunction[1] is proportional to

$$|\Psi_{\mathrm{N.B}}[h_{ij}, \phi_0]| \propto \exp\left[+\frac{12\pi^2}{\kappa^2 V(\phi_\star)}\right], \qquad \kappa = 8\pi G_N\,, \tag{1.2}$$

[1] In the context of quantum cosmology, the relevant manifold descrbing the no-boundary instanton is a complex geometry that smoothly interpolates between a Euclidean geometry and a Lorentzian geometry. The Lorentizan evolution gives a multiplicative oscillatory term.

where $\phi_\star$ is the value of the inflaton field at the turning point of the geometry, corresponding to the onset of inflation. Famously, this result is plagued by the fact that the wavefunction is non-normalizable and that it favors the smallest possible number of e-folds. This raises both a theoretical puzzle concerning how such a result should be interpreted and a phenomenological tension, since it does not explain why the universe is observed in its present state rather than being essentially empty. This has been the subject of many studies for the last 40 years. For a recent review, see [2, 3] and the references therein. In what follows, I will summurize a new paradigm that has been developed to adress the issues affected by the Hartle-Hawking proposal [4–6].

One of the central lessons of the AdS/CFT correspondence is the deep connection between spacetime geometry and quantum entanglement, most notably illustrated by the Ryu–Takayanagi formula [7–9]. Understanding the origin and generality of this connection therefore calls for studying entanglement in controlled models of quantum gravity, and in particular in string theory. This is, however, a subtle problem: because strings are extended objects, the usual notion of spatial entanglement does not admit a straightforward generalization. Motivated by this question, we have investigated entanglement entropy in (c=1) non-critical string theory, a particularly tractable setting that admits a dual description in terms of Matrix Quantum Mechanics (MQM).

In parallel, another aspect of my work concerns two-dimensional conformal field theories and Liouville theory. Away from the critical central charge, Liouville theory naturally emerges through the conformal anomaly. The bootstrap program has led to a remarkably detailed understanding of spacelike Liouville theory, including exact results for CFT data both on the sphere and on the upper half-plane. By contrast, timelike Liouville theory, with $c_L \leq 1$, remains much less understood, particularly in the presence of a boundary. Addressing this gap, and developing a better understanding of timelike Liouville theory on the upper half-plane, is the focus of our ongoing work.

Finally, the last project is a study of Einstein + axion theory in presence of periodic potential arising from non-perturbative effects. Such system can admit asymptotically AdS wineglass wormholes. We aim to study both numerically and analytically the homogeneous and isotropic solutions to Einstein equations using the tools of dynamical systems.

The report is structured as follows. In Section 2, I will review the work on wineglass wormholes and their applications to cosmology. In Section 3, I will provide an executive summary of the ongoing projects, focusing on the main results obtained so far and on the remaining open questions. I will also use this *comité de suivi* as an opportunity to assess the work that remains to be done on these projects.

2 Wineglass wormholes and cosmology

2.1 A paradigm that tends to tackle the issues encountered in the Hartle-Hawking proposal

The proposal comes from the observation that the issue afflicting the no-boundary proposal (1.2) stems from the fact that the euclidean action of an Euclidean de Sitter spacetime is negative. This sign issue is naturally tackled by considering the possibility of an euclidean preparation interpolating a region for which the potential is negative to a region where the potential is positive (see Fig. 1). This typical fine tuned shape of the potential is motivated by Higgs inflation scenario [5]. The no-boundary solution can be thought as the euclidean sphere which we cut in half and analytically continue to a Lorentzian de Sitter spacetime. The analog of such $\mathbb{Z}_2$ symmetric solutions in the case of the potential depicted in Fig.1 correspond to euclidean wormholes. For the euclidean wormholes to support interesting cosmologies without crunch, the Z_2 symmetric point should be a local maximum of the scale factor (see Fig. 2). The proposal can be incorporated in a concrete

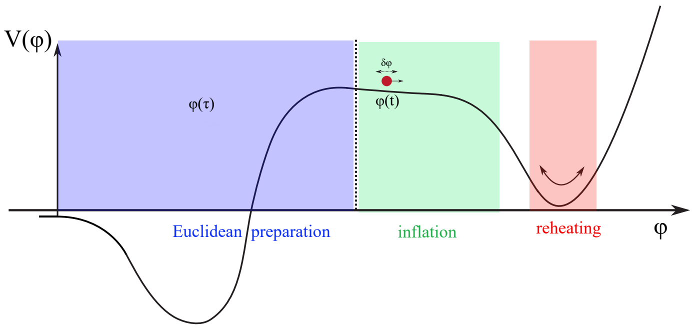


Figure 1: Schematic representation of the desired potential that leads to inflationnary cosmology from a half-wineglass euclidean geometry. Figure from [4].

Einstein-Maxwell($U(1)^3$)-dilaton $(g_{\mu\nu}, F^{(I)}_{\mu\nu}, \phi)$ model [6][2].

$$\begin{aligned} \mathcal{S}_E &= \int \mathrm{d}^4x\sqrt{g_E}\left(-\frac{1}{2\kappa}\mathcal{R} + \frac{1}{2}(\partial_\mu\phi)^2 + V(\phi) + \mathcal{L}^E_{\mathrm{rad}}\right) + \mathcal{S}_{\mathrm{GHY}}\,, \\ \mathcal{S}^E_{\mathrm{rad}} &= \int \mathrm{d}^4x\sqrt{g_E}\frac{1}{4}\sum_{I=1}^{3} F^{(I)\,\mu\nu}F^{(I)}_{\mu\nu}\,. \end{aligned} \tag{2.1}$$

The objective of the paper [6] was to study the holographic properties and the phase diagram of homogeneous and isotropic euclidean solutions as a fonction of the source parameters turned on at the EAdS boundary. Such phase diagram ultimately informs on which euclidean preparation of the state is the most plausible. In what follows, I will only focus on describing specific characteristics of the solutions and on the results. All the computational details can be found in the paper.

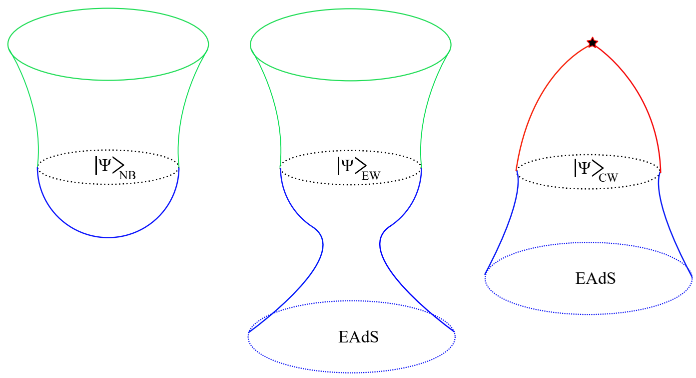


Figure 2: **Left**: the analytic continuation of the euclidean de Sitter sphere leads to the no-boundary proposal. **Center**: Analytic continuation of wineglass wormhole leading to an half-wineglass geometry. **Right**: The analytic continuation of a wormhole whose $\mathbb{Z}_2$ symmetric point is a minimum leads to a cosmological crunch. Figure from [4].

[2]Euclidean wormholes with spherical slices need to be supported by a negative euclidean energy density. This can be achieved by considering a magnetically dominated Maxwell field or an axion field for examples. Here we will focus on the first possibility. More details can be found in [10].

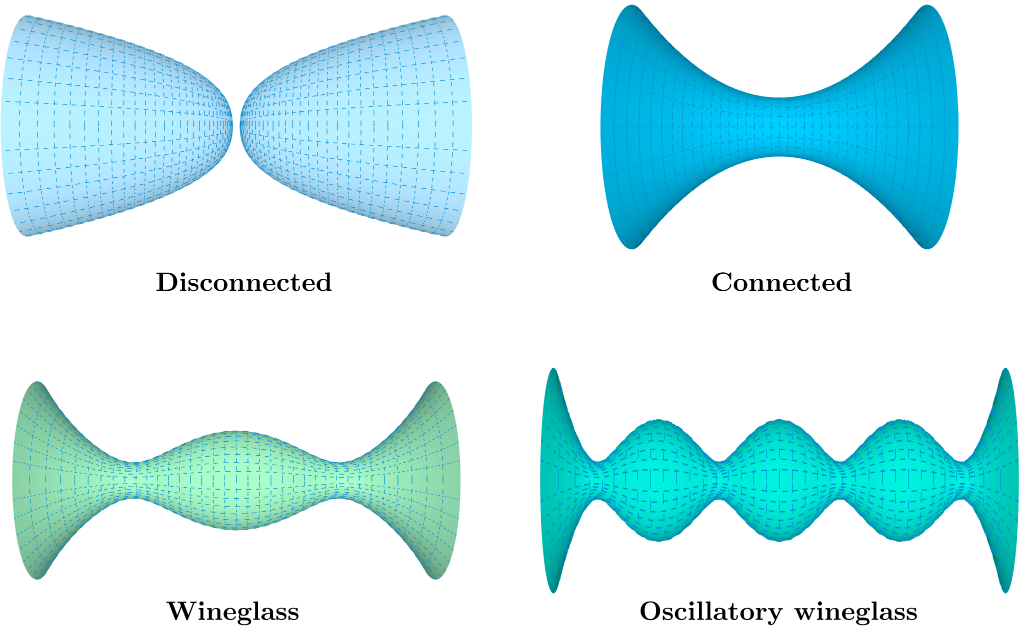


Figure 3: The different types of Euclidean AdS gravitational saddles that we analyze and compare in [6]. **Top**: non-running solutions. **Bottom**: More complicated running solutions.

2.2 Phase diagram of homogeneous and isotropic euclidean solutions in a concrete Einstein-Maxwell-dilaton model

We are looking for an isotropic and homogenous solution (ω^I denotes the Maurer-Cartan forms on S^3)

$$\mathrm{d}s^2_{\mathrm{EFLRW}} = \mathrm{d}\tau^2 + a^2(\tau)\mathrm{d}\Omega_3^2\,, \quad \phi = \phi(\tau)\,, \quad A^{(I)}(\tau) = H(\tau)\omega^I\,, \tag{2.2}$$

to the equations of motion (EOMs)

$$\begin{aligned}
\tilde{\phi}'' + 3\frac{a'\tilde{\phi}'}{a} - \frac{\mathrm{d}\tilde{V}(\phi)}{\mathrm{d}\tilde{\phi}} &= 0\,,\\
\frac{2a''}{a} + \frac{a'^2}{a^2} - \frac{1}{a^2} + 3\left(\tilde{V}(\phi) + \frac{\tilde{\phi}'^2}{2}\right) + \frac{\tilde{\rho}^E_{\mathrm{rad}}}{a^4} &= 0\,,\\
\frac{a'^2}{a^2} - \frac{1}{a^2} + \left(\tilde{V}(\phi) - \frac{\tilde{\phi}'^2}{2}\right) - \frac{\tilde{\rho}^E_{\mathrm{rad}}}{a^4} &= 0\,,\\
\tilde{\rho}^E_{\mathrm{rad}} = 2\kappa a^4\left[\left(\frac{H'}{a}\right)^2 - \frac{4H^2}{a^4}\right] &= \mathrm{const}\,.
\end{aligned} \tag{2.3}$$

For convenience, we defined rescaled quantities $\tilde{V}(\phi) = \kappa V(\phi)/3$, $\tilde{\phi} = \sqrt{\kappa/3}\phi$. There are two sets of solutions (see Fig.3) for a given potential of the form depicted in Fig.1 (we focus on the euclidean part only):

- Non-running solutions for which the field ϕ sits at the EAdS maximum. The disconnected solution is particular as it is a self-dual solution for which the euclidean energy density vanishes.

- Running solutions interpolating from the EAdS maximum to the positive region of the potential. From the holographic point of view, these solutions are characterized by a renormalization group flow driven by a relevant operator.

Let me also mention that, for wormholes, the electromagnetic energy density can be naturally expressed in terms of the electromagnetic field at the center of the wormhole H_0

$$\tilde{\rho}^E_{\text{rad}} = -8\kappa H_0^2\,. \tag{2.4}$$

This implies that holographically the electromagnetic energy density is an IR quantity and not a UV parameter of the model, so that backgrounds with different E/M density can be compared, as long as the gauge field has the same UV (source) asymptotics.

Since it is very hard to analytically solve the EOMs for a given potential that supports wineglass wormhole solutions[3], let alone to do this for a class of potentials with the wanted characteristics, we proceeded in a reverse manner in finding the solutions of interest. Remark that from the equations of motion, once the scale factor is known, eveyrything including the potential is known. We thus first proposed an analytic $\mathbb{Z}_2$ symmetric ansatz for the scale factor comprised out of three regions[4]: (**I**) for $-\infty < \tau < \tau_{\text{min}}$ and (**II**) for $\tau_{\text{min}} \leq \tau \leq -\tau_{\text{min}}$ and (**I′**) for $-\tau_{\text{min}} < \tau < \infty$ where the scale factor is similar to that in region (**I**)

$$a^2_{WG}(\tau) = \begin{cases} \dfrac{1}{4H^2_{\text{AdS}}\cosh(2H_{\text{AdS}}(\tau-\tau_{\text{min}}))}\left(\sqrt{A_{\text{AdS}}}\cosh(2H_{\text{AdS}}(\tau-\tau_{\text{min}})) + \dfrac{c_{\text{AdS}}}{2\sqrt{A_{\text{AdS}}}}\right)^2\,, & (\mathbf{I}) \\ \dfrac{1}{4H^2_{\text{dS}}}\left(\cos\left(2H_{\text{dS}}\tau\right) + c_{\text{dS}}\right)\,, & (\mathbf{II}) \\ \dfrac{1}{4H^2_{\text{AdS}}\cosh\left(2H_{\text{AdS}}(\tau+\tau_{\text{min}})\right)}\left(\sqrt{A_{\text{AdS}}}\cosh\left(2H_{\text{AdS}}(\tau+\tau_{\text{min}})\right) + \dfrac{c_{\text{AdS}}}{2\sqrt{A_{\text{AdS}}}}\right)^2\,. & (\mathbf{I'}) \end{cases} \tag{2.5}$$

with $A_{\text{AdS}}, H_{(\text{A})\text{dS}}, c_{\text{dS}} > 0$. We also have $\tau_{\text{min}} = -\pi/2H_{\text{dS}}$, when the scale factor reaches its minimum size. The parameters $H_{(\text{A})\text{dS}}$ have dimensions of mass, while all other quantities are dimensionless. Few remarks are in order concerning these wineglass solutions:

- The running solutions are characterized by a scalar field with conformal dimension $\Delta_+ = 2$. c_{AdS} is a parameter that drives the presence of a source for the scalar field (in Dirichlet boundary condition). If $c_{AdS} \neq 2$, the source of the scalar field is turned off.

- There are 5 free parameters in this ansatz but three constraints. Two are obtained by asking the scale factor to be $\mathcal{C}^2$ and the other one is obtained by requiring the velocity of the scalar field to vanish. This condition relates the parameters to the euclidean energy density. We effectively have a model with one additional parameter compared to the non-running solutions which only have H_{AdS} and the euclidean energy density.

As it is well known, for asymptotically (E)AdS spacetimes, the on-shell action is divergent and one has to renormalize it in order to obtain a final result. In our case, we decided to implement background substraction computing the difference of on-shell actions after regularizing them (the procedure is explained in section 3.2 of our paper). In the paper, we analyse both Dirichlet or Neumann boundary conditions (BCs) for the scalar field (we also analysed Dirichlet or Neumann BCs for the gauge field). Some results are displayed in Fig.4. As can be seen, the wineglass

[3]Concerning oscilatory wineglass solutions, there are easily obtain by duplicating the central region of the potential.

[4]In the paper, a more general ansatz was proposed but I decided to restrict myself to the case where everything is analytically solvable. Once the scale factor is known the potential can be inferred and is presented in Appendix B of the paper.

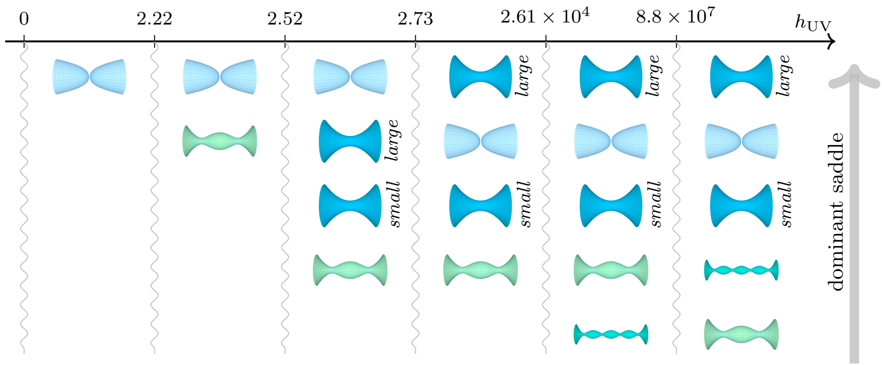


Figure 4: Dominant saddles in the gravitational path integral across a range of $h_{\rm UV} \equiv (\kappa|\tilde{V}_{\rm AdS}|)^{1/2} H_{\rm UV}$ for $A_{AdS} = 10^6$. Dirichlet BCs are imposed for the gauge field and Neumann BCS for the scalar field. Some saddles cease to exist below certain minimum values $h_{\rm UV}^{\rm min}$. The labels *large* and *small* denote the two connected wormhole branches. Figure from [6].

wormhole can become the dominant wormhole contribution. In the paper, it was argued that the disconnected solution should be discarded in discussing the probability of nucleating a universe with certain conditions at the start of inflation. Upon this assumption, we thus conclude that depending on the values of the free parameters, the nucleation of the universe can be explained by a certain wineglass shape geometry in this model.

2.3 Inclusion of perturbations

The work achieved so far is a zeroth order discussion which concerns the possibility of having an euclidean preperation corresponding to a half-wineglass geometry. Cosmology is about the fluctuations of the background. The natural question to ask is whether cosmological correlators in the wineglass background are the same as in the no-boundary background[5]. The most direct way to settle such question is through the wave-function formalism [11–13]. Once the wavefunction is known, all equal-time correlators can be derived. Suppose we consider a free massless scalar probe Φ on top of the background. Since we are dealing with a free theory, the path-integral is Gaussian and reduces to a boundary term on the spacelike hypersurface. As previously mentioned, the two constructions differ primarily in the Euclidean geometry used to prepare the state and in the condition imposed at its initial end. In the no-boundary geometry, regularity is imposed at the south pole which we set at $\tau_i = -\pi/2H$

$$\Psi_{N.B}[\Phi(t_f,\Omega)] = \mathcal{N} \exp\left[i \int \mathrm{d}\Omega \frac{\mathrm{a}^3_{\rm NB}(\mathrm{t_f})}{2} \Phi_{\rm cl}(\mathrm{t_f},\Omega)\partial_{\rm t}\Phi_{\rm cl}(\mathrm{t_f},\Omega)\right] . \tag{2.6}$$

Here the subscript "cl" indicates that $\Phi_{\rm cl}$ should be a solution of the classical Klein-Gordon equations of motion while $a_{NB} = \frac{1}{H}\cos(H\tau) = \frac{1}{H}\cosh(Ht)$ denotes the no-boundary background scale factor. It is well known that such path-integral prepares the Bunch-Davies vacuum in de Sitter space. In the wineglass geometry the geometry asymptotes to EAdS and the scale factor $a_{WG}(\tau)$

[5]Here, we do not ask about the euclidean stability of such solutions which is also a crucial point. However, discussing possible negative modes is very difficult question (if not impossible) in view of the difficulty of such solutions. In the meantime, there is also a notion of stability arising in cosmological correlators which has to do with the positivity of the power spectrum (equal time two point function). It would be interesting to understand the relation between the two notions of stability.

diverges there. For the path-integral to be well defined, this selects the normalizable mode solution. The wave-function has formally the same expression as in the no-boundary case

$$\Psi_{WG}[\Phi(t_f,\Omega)] = \mathcal{N} \exp\left[i\int \mathrm{d}\Omega \frac{\mathrm{a}^3_{\mathrm{WG}}(\mathrm{t_f})}{2}\Phi_{\mathrm{cl}}(\mathrm{t_f},\Omega)\partial_{\mathrm{t}}\Phi_{\mathrm{cl}}(\mathrm{t_f},\Omega)\right]\,. \tag{2.7}$$

It remains to solve the Klein-Gordon equation for Φ for each background to compute the wave-function. In the case of the no-boundary proposal, exact results are known [2, 3]. In the case of the wineglass proposal, we have to resort to numerical tecniques to obtain the two-point function[6]. This was achieved in [14] for specific wineglass solutions. The results are that the power spectrum differs only for small values of the harmonic numbe of the 3-sphere. We do not expect to be able to see such effects in the CMB data because probing such effect is tantamount to probing the size of the curvature of the universe at the end of inflation. This is a striking conclusion that two different euclidean preparations of the state, although geometrically very different can not be disguished in the data!

Future directions: Find the power spectrum numerically and possibly analytically using a WKB matching procedure. **Distance to the end**: 3 to 6 months.

3 Updates of the current projects

3.1 Entanglement entropy in $c = 1$ non critical string theory

The $c = 1$ model is originally a sigma model which consists in one boson X. Upon gauge fixing, the model divides into 3 sectors Liouville/ghost/matter

$$\mathcal{S}_{X+L} = \frac{1}{4\pi}\int \mathrm{d}^2\mathrm{z}\sqrt{\hat{\mathrm{g}}}\left(\hat{\mathrm{g}}^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + 2\hat{\mathcal{R}}\phi + 4\pi\mu \mathrm{e}^{2\phi} + \hat{\mathrm{g}}^{\mu\nu}\partial_\mu\mathrm{X}\partial_\nu\mathrm{X}\right)\,. \tag{3.1}$$

It is well known that this model is dual to the double scaled limit of a Matrix Quantum mechanics [15–17]. The model has also a very simple interpretation from the target space perspective describing a 2D target space spans by the dilaton and the boson (X,Φ) with $\Phi \propto 2\phi$ (the string coupling is $g_s = e^{\Phi}$). Hence, this duality offers a simple way to analyse the emergence of spacetime arising from the matrix degrees of freedom. Entanglement entropy is a very nice diagnose of the emergence of spacetime from the matrix degrees of freedom. The question is whether the EE entropy of a subregion $\mathcal{R}$ can reproduced the generalized entropy of the gravitational theory.

$$S_{\mathrm{gen}} = \frac{A_{\partial\mathcal{R}}}{4G_N} + S_{\mathrm{mat}}(\mathcal{R})\,. \tag{3.2}$$

In the case of 2D gravitational theory, it is expected that the so-called "area-terms" are proportional to $\frac{1}{g_s^2} = e^{-4\phi}$ [18]. The beginning of the story is a computation achieved by Hartnoll and Mazenc in 2015 [19]. They computed the entanglement entropy of a cut in the eigenvalue variable $A = [\lambda_1, \lambda_2]$ finding

$$S_A = \frac{1}{3}\log\left(\frac{\lambda_2-\lambda_1}{\sqrt{g_s(\lambda_1)g_s(\lambda_2)}}\right) + \dots\,. \tag{3.3}$$

This formula reproduces the Cardy-Calabrese universe law of a 2d conformal field theory of central charge $c = 1$ [20]. This computation was subsequently confirmed in 2023 using the replica trick in the hydrodynamical effective theory for the eigenvalues [21]. No area-terms were found. According to a conjecture by Susking-Uglum [22], the entanglement entropy across the horizon of a black hole

[6]The equations of motion in the no-boundary case are hypergeometric while in the case of the wineglass proposal, these are Fuchsian equations of higher degree. I believe that we could find analytic results in the case of large harmonic number using WKB matching procedure methods.

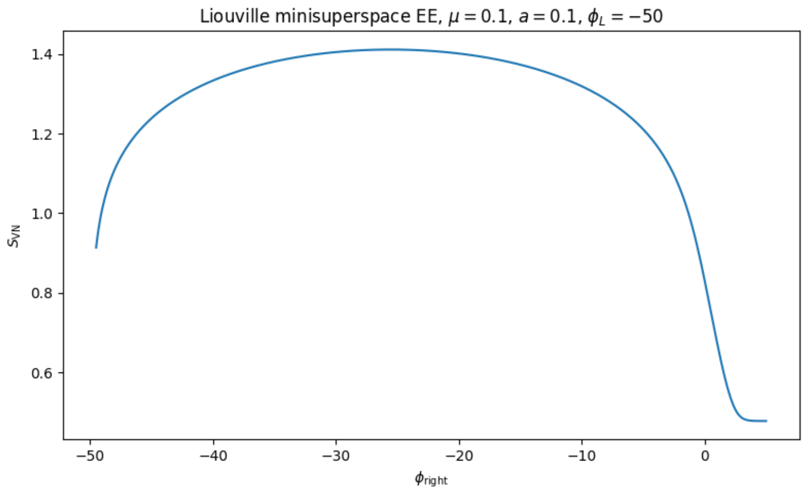


Figure 5: Entanglement entropy as a function of the $\phi_{\rm right}$.

in string theory is due to the endpoints of open strings anchored to the horizon. This was thus argued that such area terms will emerge from the non-singlet sector of Matrix Quantum mechanics which is dual to two dimensional string theory in the black hole background [23, 24].

The aim of our work is to explore all of these questions. The first remark regarding the previous computation is that all of them were performed from the matrix model degrees of freedom which are then subsequently related to the target space coordinates (X, Φ); the eigenvalue λ is related to the field ϕ by a non local transform and only in the weakly coupled region such relation simplifies. To access results in the strongly coupled region and potential effects arising near the Liouville wall, we would like to perform a computation directly in target space coordinates. The direction taken is widely inspired by works achieved Balasubramanian and Parrikar in 2018 [25] where they proposed computing Entanglement entropy for spatial subregions in bosonic string theory using the framework of string field theory. In the case of $c = 1$ non critical string, the string field action can be derived from the minisuperspace Wheeler-De-Wiit constraint which reveals the propogator

$$\mathcal{S}_{\rm free} = \frac{1}{2}\int {\rm d}\phi {\rm dX} \left[(\partial_\phi \Psi)^2 + (\partial_{\rm X}\Psi)^2 + 4\pi\mu {\rm e}^{2\phi}\Psi^2 \right] . \tag{3.4}$$

This effectively describes a 1+1 field theory ($X = it$) which is not translation invariant. For the sake of simplicity, I will summurize the progresses made and the future directions:

- We have both numerical procedures (mapping the continumm theory to a lattice theory) (see Fig. 5) and we also aim to compare these to analytic calculations using the replica trick.

- Are there really no area-terms ? It seems that there might be an exponential suppression near the Liouville wall but we are not sure yet if these are $e^{-4\phi}$ or not (see Fig. 5).

- Can we extend the calculation to the minisuperspace string field action (3.4) ? The matrix model should provide a way to give corrections to the propagator [26].

- Can we easily extend such computations (at least numerically) to the non-singlet sector and find area terms ?

3.2 The bootstrap program of Timelike Liouville theory

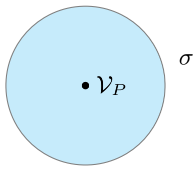
Notations

- Central charge: $c = 1 + 6Q^2 = 1 + 6(b + b^{-1})^2$
- Bulk conformal dimensions: $\Delta = \frac{Q^2}{4} - P^2$ (same for boundary conformal dimensions with additional tildes)
- The parameter σ labels FZZT boundary conditions [27] $\cosh^2(\pi b \sigma) = \frac{\mu_B}{\mu} \sin(\pi b^2)$.

The bootstrap program of boundary Liouville field theory consists in deriving the following CFT data:

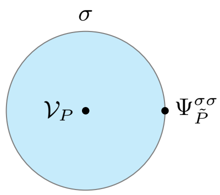
- The disk one-point function

$$\langle \mathcal{V}_P(z) \rangle_{\mu_B} = \frac{U(P|\sigma)}{|z - \bar{z}|^{2\Delta}} = \tag{3.5}$$



- The bulk-boundary two-point function

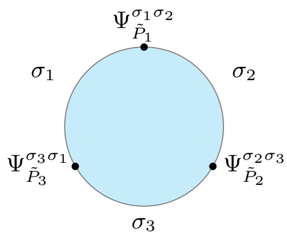
$$\langle \mathcal{V}_P(z)\, \Psi^{\sigma\sigma}_{\tilde{P}}(x) \rangle = \frac{R(P, \tilde{P}|\sigma)}{|z - \bar{z}|^{2\Delta - \tilde{\Delta}} |z - x|^{2\tilde{\Delta}}} \tag{3.6}$$



- The bulk three-point function

$$\langle \Psi^{\sigma_1\sigma_2}_{\tilde{P}_1}(x_1) \Psi^{\sigma_2\sigma_3}_{\tilde{P}_2}(x_2) \Psi^{\sigma_3\sigma_1}_{\tilde{P}_3}(x_3) \rangle = \frac{C^{\sigma_1\sigma_2\sigma_3}_{\tilde{P}_1\tilde{P}_2\tilde{P}_3}}{|x_{12}|^{\tilde{\Delta}_1 + \tilde{\Delta}_2 - \tilde{\Delta}_3} |x_{23}|^{\tilde{\Delta}_2 + \tilde{\Delta}_3 - \tilde{\Delta}_1} |x_{13}|^{\tilde{\Delta}_1 + \tilde{\Delta}_3 - \tilde{\Delta}_2}}$$

$$= \tag{3.7}$$



In the case of spacelike Liouville $c \geq 25$, this CFT data was obtained by Fadeev, Zamolodchikov, Zamolodchikov, Ponsot and Teschner [27–29]. The CFT data in the Timelike case is still a curent ongoing topic of research related to how analytically continue the results found by the previous authors [30, 31]. The approach we are advocating with Ioannis Tsiares is based on the works achieved by Ponsot and Teschner [29]. They remarked that the solution of the Pentagon equation provided by the fusion kernels [32–34] yield a solution for the crossing symmetry equation satisfied by the boundary three point function. The knowledge of the boundary Liouville data is thus tied to the knowledge of the crossing kernels[7]. In the meantime, recent works by Tsiares-Roussilon [35]

[7] There is a remarkable relation between crossing symmetry of 2 bulk and one boundary correlator and a Moore-Seiberg consistency condition of the torus two point function [32]. This relation can be exploited to find the bulk to boundary CFT data. This is a crucial observation of our coming works with Ioannis Tsiares.

conjectured a form of the Timelike crossing kernels for rational $\hat{b} = (ib)^2$. We are now trying to check if these kernels (which are non-meromorphic!) satisfy equivalent Moore-Seiberg consistency conditions which would provide a definite solution for the CFT data.

3.3 Axion wormholes and Holography with the tools of dynamical system

We recently got interested in studying axion wormholes[8] [37, 38] lifted by non-perturbative potential [39, 40]

$$\mathcal{S} = \int d^D x\sqrt{g}\left[\frac{1}{2\kappa}\left(-\mathcal{R}+2\Lambda\right) - \frac{1}{2}\partial_\mu\chi\partial^\mu\chi + V_{\rm np}(\chi)\right] + \mathcal{S}_{GHY} - \int \chi\wedge\star d\chi\,. \tag{3.8}$$

There are multiple reasons for that. First, in asymptotically dS spacetime, there is a paradox that in absence of a potential, there is no physical principle preventing the number of bounces to be infinite [39]. Second, depending on the values of the parameters, if we have a effective potential for the axion which admits both positive and negative extrema, the system can support wineglass wormholes. I've been exploring the phase space of homogeneous and isotropic solutions using the tools of dynamical system[9] for potentials supporting wineglass type of solutions.

$$V_{np}(\chi) = V_0 + V_1\cos\left(\frac{\chi}{f_\chi}\right)\,, \quad V_1 > V_0\,. \tag{3.9}$$

For homogeneous and isotropic solutions in spherical slices in $D = d+1$ dimensions,

$$\mathrm{ds}^2 = \mathrm{d}\tau^2 + \mathrm{a}^2(\tau)\mathrm{d}\Omega_\mathrm{d}^2\,, \quad \chi = \chi(\tau)\,, \tag{3.10}$$

the equations of motion are

$$\begin{aligned} \partial_\tau\left(a^d\partial_\tau\chi\right) + a^d\partial_\chi V_{np}(\chi) &= 0\,. \\ \frac{d(d-1)}{a^2}\left(a'^2-1\right) + 2\kappa\left(\frac{(\partial_\tau\chi)^2}{2} + V_{\rm np}(\chi) + \frac{\Lambda}{\kappa}\right) &= 0\,, \\ \frac{(d-1)(d-2)}{a^2}\left(1+a'^2\right) - \frac{2(d-1)}{a^{d-1}}\frac{d}{d\tau}\left(a'a^{d-2}\right) + 2\kappa\left(\frac{(\partial_\tau\chi)^2}{2} - V_{\rm np}(\chi) - \frac{\Lambda}{\kappa}\right) &= 0\,. \end{aligned} \tag{3.11}$$

We can rewrite this set of equations as a first order system of dimension 3. Introducing dimensionless quantities $v_0 = \frac{V_0}{\kappa V_1}$, $\alpha = f_\chi\sqrt{\kappa}$ and dimensionless vairbales $\mathcal{I}\propto\chi$, $\mathcal{J}\propto\chi'$, $\mathcal{H}\propto H$, we obtain

$$\begin{aligned} \frac{d\mathcal{J}}{du} &= -d\mathcal{J}\mathcal{H} + \frac{\sin(I)}{\alpha}\,, \\ \frac{dI}{du} &= \frac{\mathcal{J}}{\alpha}\,, \\ \frac{d\mathcal{H}}{du} &= -\mathcal{H}^2 + \frac{\mathcal{J}^2}{d} - \frac{2}{d(d-1)}\left[v_0 + \cos(I)\right]\,. \end{aligned} \tag{3.12}$$

Among the remarks and the results that I want to mention are the following

- Because of the non-perturbative potential, the axion charge $Q \sim a^d\partial\tau\chi$ is no longer a constant of motion.
- The action for χ yields a Neumann like problem. In absence of a potential, the charge is related to the source (finiteness of the charge imposes the vev of the field χ to vanish).

[8] The axion system is originally a $(D-1)$-form system. Upon dualization, this gives rise to a system with a scalar adimitting a negative kinetic term. See [36] for more details.

[9] Fortunately, such dynamical system methods have been widely developed by cosmologist but in Lorenztian signature [41–43].

- From the dynamical system point of view, stable critical points correspond to EAdS fixed points. Depending on the value of the "conformal dimension" (found by expanding the field near the EAdS fixed point) Δ of the axion field, the trajectories are either spiralling or not. Spiralling trajectories correspond to conformal dimensions which break the BF bound [44, 45].
- Any non trivial running solutions satisfying the BF bound carry infinite axionic flux at the EAdS boundary.

4 Conclusion and acknowledgements

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